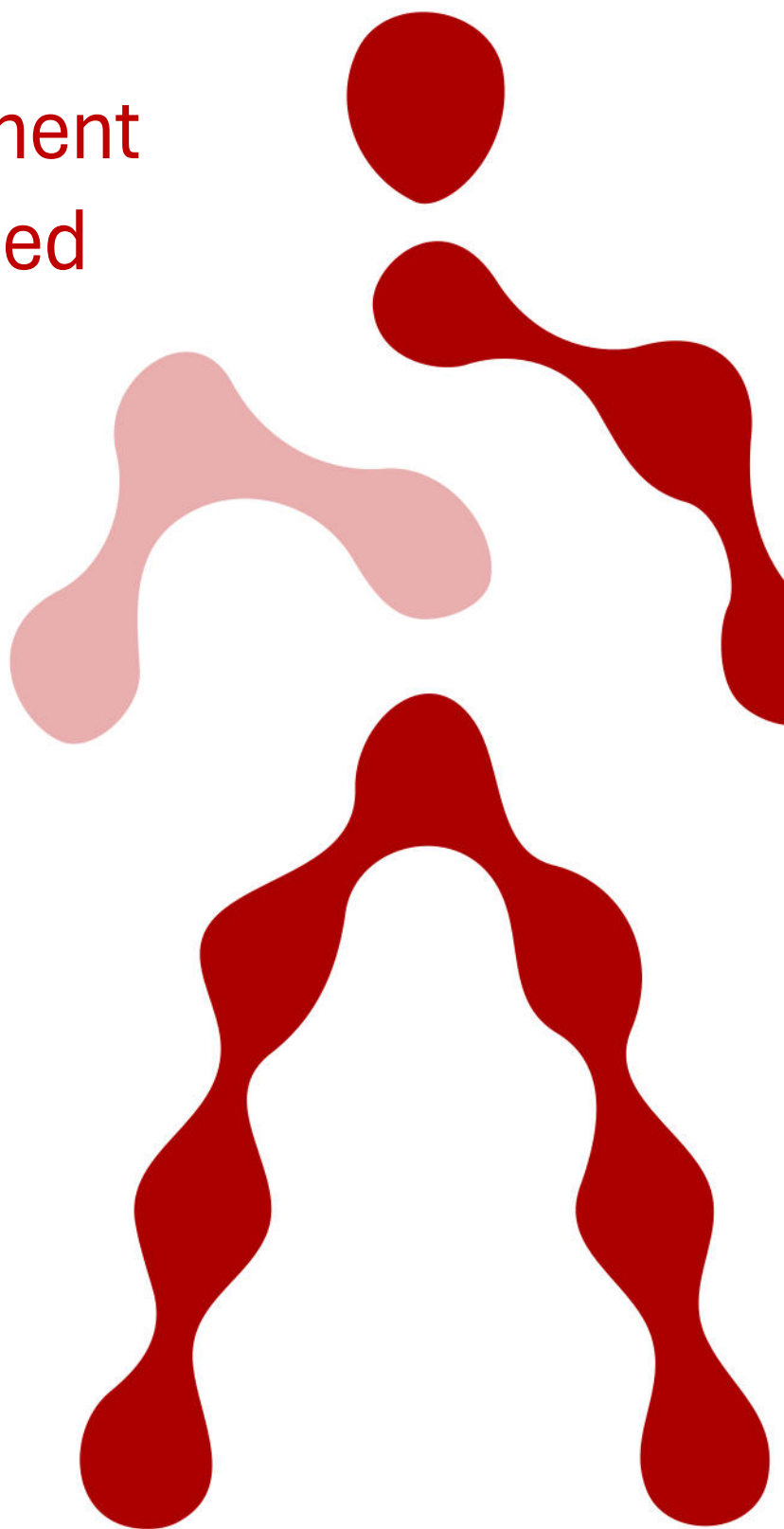


Initial clinical management of patients exposed to weaponized chemicals

Interim guidance

April 2026



World Health
Organization

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Introduction

This guidance is aimed at health care workers who may receive patients exposed to chemical weapons or other hazardous or toxic chemicals at their health care facilities.

The guidance follows the clinical management flowchart in Figure 1.

It provides questions to guide the identification of contaminated patients, recommendations on personal protection, procedures for decontamination, guidance for triage and identification of categories of exposure, and treatment regimens for individual chemicals.

Users should study the contents of this document carefully and apply the principles and framework to their own situation and health care facilities.

Clinical work in this field should be accompanied by complete and practical training.

Initial management of patients

Figure 1: Flowchart describing the initial management of patients



Decide whether patient has been exposed to chemicals

Key principles

Treatment of life-threatening conditions is the priority, as soon as it is safe to do so, and even before considering decontamination. Patients may have been exposed to chemicals through inhalation, contact with the skin or eyes, by ingestion or by a contaminated projectile e.g. shrapnel. The priorities for treatment adopt an all-hazards sequential approach: ^[1, 2]

- Catastrophic haemorrhage
- Airway (and specific antidote where available)
- Breathing
- Circulation
- Decontamination (and disability)
- Evacuation to a more permissive environment

In the case of suspected chemical exposure, the use of “quick look” assessment is used, following the acronym CRESS (Consciousness, Respiration, Eyes, Secretions, Skin).^[3] Remember that mixed chemical exposures can occur. Injuries from trauma and other medical complications may also be present.

Life saving interventions can be performed before decontamination by any appropriately trained person when the immediate threat to a patient’s life, for example from catastrophic hemorrhage or cardiorespiratory arrest is higher than ongoing risk from the contamination itself. Depending on the resources and competencies available, medical personnel must decide in advance what level of resuscitation could be attempted before, or in, the decontamination area, as well as the criteria for resuscitation.^[4]

Decontamination

- Exposure to a gas, vapour or aerosol may not leave visible signs of contamination but can still require decontamination.
- Patients presenting at health care facilities following exposure to chemical attacks should be considered by default as contaminated and require urgent decontamination.
- Patient decontamination should occur in combination with triage and the provision of life-saving interventions.

Table 1: Screening of patients for weaponized chemical exposure before entering the health care facility

Before the patient enters the health care facility, ask the following questions	
1 What is the history of exposure of the patient?	
A	Where was the patient? When did they start experiencing symptoms? What did they experience first? Were others experiencing similar symptoms?
B	Did the patient notice an unusual smell, e.g. garlic (may indicate mustard gas), bitter almonds (may indicate cyanide), fresh hay or grass (may indicate phosgene).
C	Take a family/patient/witness/first responder report.
D	Use contextual information (e.g. health authorities, law enforcement, reputable media sources etc.)
2 Can you observe any signs of chemicals on or around the patient?	
A	Dust, powder or liquid droplets on body surface or clothes.
B	Discoloration of clothes, scorching or damage to clothing (e.g. indicating a chemical reaction).
C	Non-exposed persons accompanying the patient show symptoms and signs suggesting secondary exposure.
D	If available, chemical agent detection procedures/equipment (e.g. a chemical agent monitor, rapid test for sulfur mustard or cholinesterase activity) should be utilized
3 Are there signs and symptoms of exposure?	
Does the patient appear unwell? See list of signs and symptoms in Table 3.	
If there is any suspicion that the patient is contaminated with chemicals, decontamination is an immediate priority. Contaminated clothing should be removed as soon as possible and discarded appropriately as chemical waste.	

Protect yourself and other personnel

Personal Protective Equipment (PPE) is essential to first responders and personnel responsible for decontamination, triage and emergency treatment at the health care facility.

Key principles

- Health care workers are mainly exposed to toxic chemicals through direct contact with the agent on patients' skin/clothing or by inhalation or mucosal contact with a vapour hazard.
- The effective use of PPE is dependent on availability, training and an understanding of the mechanisms of secondary exposure.
- Appropriate standards and levels of PPE are dependent on the resources at different health care facilities and on the properties of the chemical agent.
- All PPE confers a loss of mobility, dexterity, vision and ability to communicate freely. PPE also places an increased physiological burden on the user.
- Even at higher levels of protection, PPE does not completely eliminate the risk of agent penetration due to eventual break-down in protective barriers.
- Remove PPE carefully in a designated location and dispose of as hazardous waste. Avoid touching any part of the clothing that may be contaminated.

General principles should be employed to provide a minimum level of staff protection before health care workers decontaminate and treat exposed patients.

Table 2: Hazards related to contact or vapour exposure to weaponized chemicals

Contact hazard	Vapour hazard
The main contact hazard can be prevented by wearing appropriate gloves (nitrile or butyl rubber, not latex). The number and thickness of the gloves used will depend on the dexterity required by the user. The minimum for protection being two layers.^[5, 6] Chemically-resistant clothing should also be worn if available. If not available, then as a minimum disposable, fluid-resistant clothing or gowns should be utilized and regularly changed.	Standard medical and surgical masks offer no respiratory or mucous membrane protection from toxic vapours. An air-purifying respirator, e.g. with an activated charcoal filter, or self-contained breathing apparatus is required. Respirators require training, safety testing and fit testing. They can only be worn by users for limited time periods.

Prepare for emergency decontamination at health care facility

Decontamination is the reduction or removal of toxic agents to minimize ongoing exposure to the patient, to prevent contamination and secondary exposures in the health care facility. This includes physical removal, and/or chemical inactivation of the agent.

Decontamination is usually first carried out in the vicinity of the incident scene by emergency services/first responders, and then later at the health care facility before admission. In some cases, patients might be taken directly to a health care facility, without prior decontamination. This chapter covers both scenarios, where the main purpose of the decontamination is to protect both the patient and others in the health care facility.

Ensure that a decontamination area is marked and cordoned off, with single points for entry/exit, and that all people leaving the area have been decontaminated. Security personnel should be assigned to the area for crowd control and to ensure appropriate flow of individuals into /out of the decontamination area. These personnel should wear appropriate PPE.

Patient decontamination starts with full removal of the visible contamination and of all clothing. Methods for decontamination can be adapted to the situation and the resources available. These methods are broadly divided into 'dry' (using no, or only absorbent decontamination agents), or 'wet' (using water or solution-based decontamination agents). Viscous or oily agents may be difficult to remove by one method alone.

Key principles

- Exposed patients should be decontaminated outside, prior to entry into health care facilities, even if they are not displaying symptoms.
- Removal of clothing is a highly effective method of decontamination.
- Decontamination should be implemented/supervised by appropriately trained personnel wearing adequate PPE.
- In general, dry decontamination should be used for decontamination unless there are clear signs of skin irritation or burns. In general, wet decontamination should be used for contamination with caustic chemicals. Methods of decontamination, whether 'dry' or 'wet' should be adapted to local resources and the situation.
- Decontamination should occur in parallel with triage and the provision of life-saving interventions.
- Contaminated waste or clothing should be disposed of safely.
- Reasonable steps should be taken to contain contaminated water used for decontamination.

Basic equipment for emergency “blot, disrobe; rinse, dry” decontamination.

- Scissors
- Buckets (5-10 litre capacity)
- Sponges/soft brushes/washcloths
- Clean water source (ideally lukewarm) water/hosepipe for most rinsing; saline solution for wound irrigation, eyes and other mucous membranes
- Liquid soap/washing up liquid/shampoo without conditioner for dissolving the chemicals
- Disposable towels/drying cloths
- Sturdy containers for used decontamination equipment
- Large sealable plastic bags (for clothing and contaminated solid waste)
- Small sealable clear plastic bags, for personal belongings
- ID/Triage labels/tags/pens
- Replacement clothing or sheets/blankets
- Stretchers

Approach to the affected person during decontamination

Practical challenges associated with decontamination include:

- Maintaining privacy and dignity of casualties.
- Preventing hypothermia: Particularly but not only in colder climates, and the provision of replacement garments.
- Communication: can be challenging in noisy and stressful situations - particularly for those with pre-existing speech, language or communication issues. Ideally keep children with parents or carers, and consider appropriate adjustments for people depending on their needs.
- Mobility: self-decontamination can be challenging for those with disabilities (or clinical effects related to exposure). Wet decontamination and the resultant slippery surfaces can result in challenges for those with poorer mobility.

Decontamination using the blot, disrobe, rinse, dry technique

Emergency wet decontamination using the “Blot, disrobe, rinse, dry” technique is simple, effective and requires minimal equipment and training. This technique may be adapted to the situation and available resources.^[7–9]

Specialized active decontamination solutions (e.g. rapid skin decontamination lotion) may be used if available.^[10]

Contain all solid waste and water run-off from the decontamination process, where possible. This is important for preventing secondary contamination.

Step 1 – Blot

- Blot off any liquid on the skin with clean absorbent material e.g. a wound dressing or incontinence pad.
- Gently brush off any solids, e.g. powder.
- Wipe affected areas gently but thoroughly with sponge or soft brush or washcloth: this removes organic chemicals and petrochemicals (not water soluble).
- Sponges and washcloths must be replaced for every patient.

Step 2 – Disrobe

Remove contaminated clothing as soon as possible. This significantly reduces contamination.

- Explain what you are going to do before you start and as you go along.
- Do NOT pull clothing off over the head.
- Remove clothing gently and use scissors to cut clothes where necessary, avoiding sensitive or wounded body areas. Lift clothes carefully. Remove shoes as they may hold contaminated particles.
 - If clothing is adherent to the patient’s body, do not rip, pull or tear. Soak gently and thoroughly with water until clothing can be separated from underlying tissue.
- Remove all accessories: jewelry, watches, rings, hearing aids, contact lenses and seal them in a bag.
- Place clothing and accessories in a large plastic bag and label as hazardous. Where resources permit, double bagging is preferable.
- If unconscious, lift the person from the cut-off clothes to a clean stretcher and blanket.

Step 3 – Rinse (if using ‘wet’ decontamination technique)

- Gently rinse/wash affected areas with soapy water (0.9% saline for open wounds): this dilutes the contaminant and removes particles and hydrophilic chemicals.
- Start with face/airways first and work down to toes. Pay special attention to skin folds, skin creases, nails, ears, and hair.
- Flush eyes copiously with 0.9% saline as needed. Use copious amounts of water.

Step 4 – Dry

- Gently dry clean areas with disposable towels. Consider dressing open wounds.

Step 5 – Clean the environment

- Transfer patient to a new stretcher in the clean zone and thoroughly wash ‘dirty’ stretcher for future use.
- Make sure all staff self-decontaminate before leaving the decontamination area. This may require a change of clothing, so additional clothes should be available for staff.
- In later care of the patient, consider any debris removed when treating trauma injuries as contaminated.

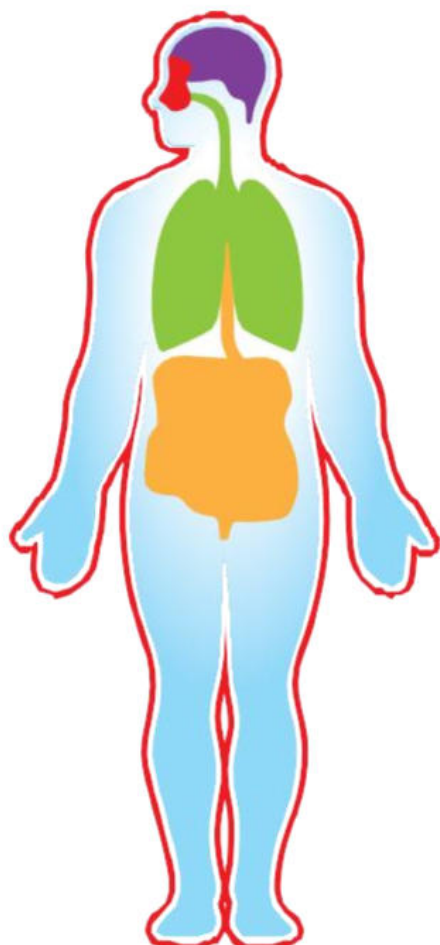
Triage and identify class of exposure

Triage activities should be adapted to the resources available at the health care facility and the scale/severity of the event. It is a dynamic process, that frequently requires repeat assessments and categorization. The flow charts on the following pages are applicable to a mass casualty event, and to resource-limited settings. Flow chart vital signs are based on adult and child parameters.

The clinical signs shown below provide a guide to the type of chemical possibly involved. Follow an initial abbreviated assessment: Quick History, Quick Look, Quick CRESS (Consciousness, Respiration, Eyes, Secretions, Skin).^[4]

The full list of signs and symptoms should be consulted for further clinical diagnosis (see below). Note that individual patients may present differently. Respiratory symptoms and signs may be present following exposure to any of the agents. Poisoning with an organophosphorus compound (e.g. nerve agent) can be confirmed on-site by measurement of reduced red-cell cholinesterase activity using a rapid-test kit.^[11]

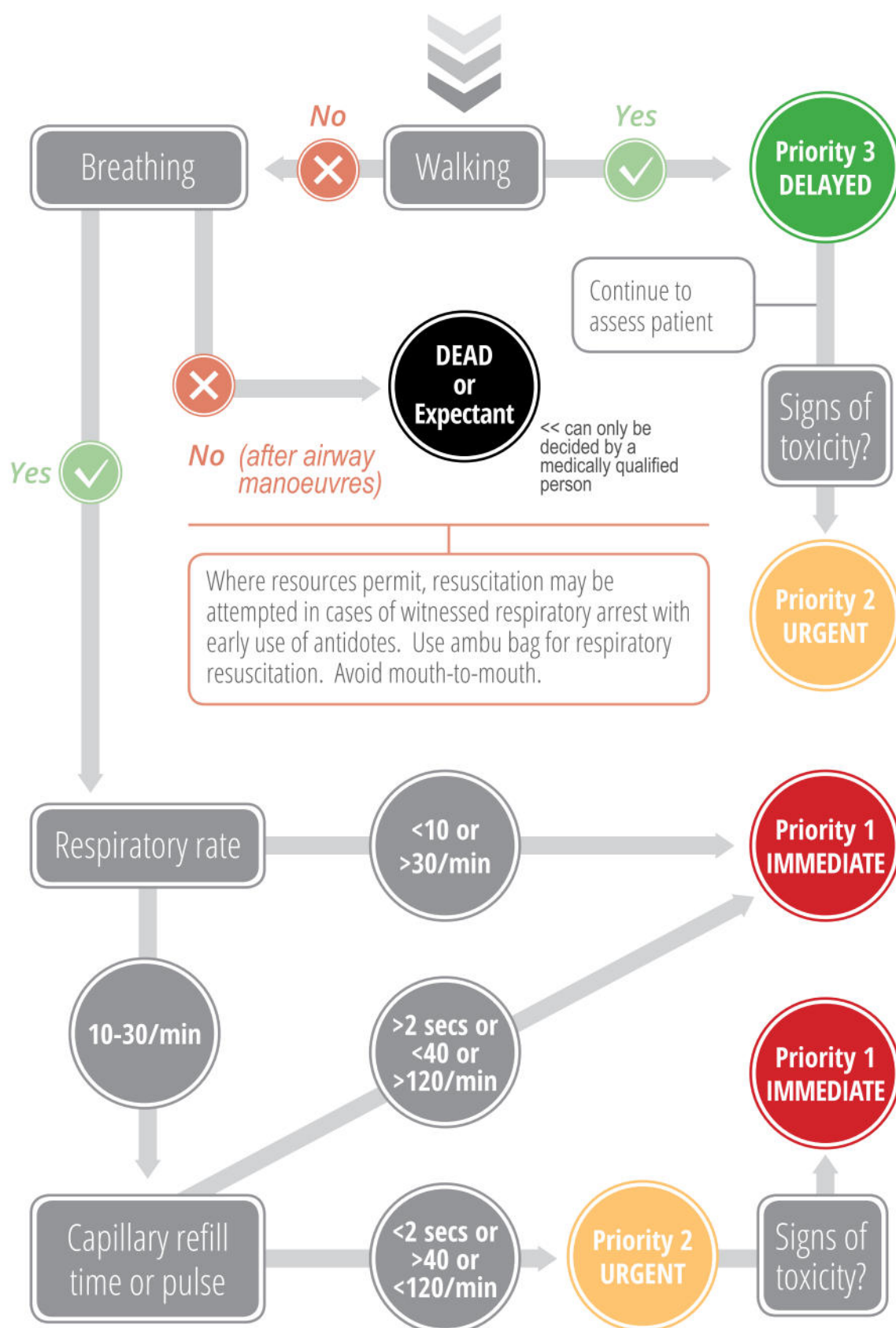
Figure 2: Symptoms associated with weaponized chemicals by organ system affected



Central nervous system	
Seizures	Cyanide, nerve agents
Hyperthermia	BZ, Agent 15
Eye, nose and skin	
Constricted pupils	Nerve agent, opioids
Dilated pupils	Nerve agent, cyanide
Dry mouth and skin	BZ, Agent 15
Eye irritation	Blister agents Riot control agents Lung irritants Phosgene
Blistering of skin	Blister agents
Cyanosis	Cyanide
Perforating thermal and chemical burns	White phosphorus
Respiratory tract	
Asphyxiation	Cyanide
Copious secretions	Nerve agents
Delayed respiratory distress	Blister agents
Delayed pulmonary oedema	Lung irritants
Irritation of the respiratory tract	Mustard gas, phosgene, riot control agents, chlorine
Digestive tract	
Nausea	Lung irritants Riot control agents Incapacitating agents Cyanide
Diarrhoea	Nerve agents
Musculoskeletal	
Fasciculations	Nerve agents

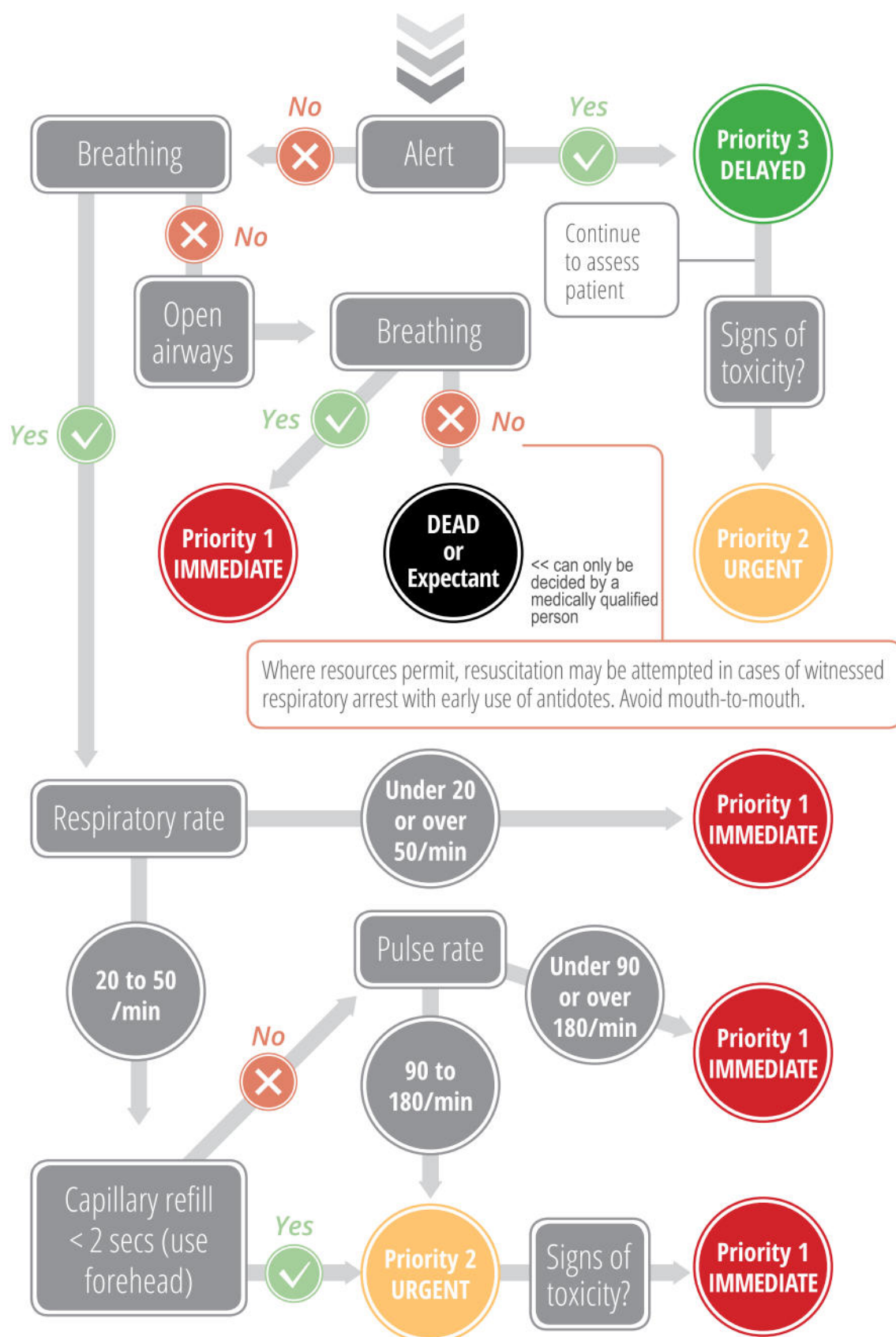
Triage: Adults

Figure 3: Flowchart of the triage of adults suspected of exposure to weaponized chemicals



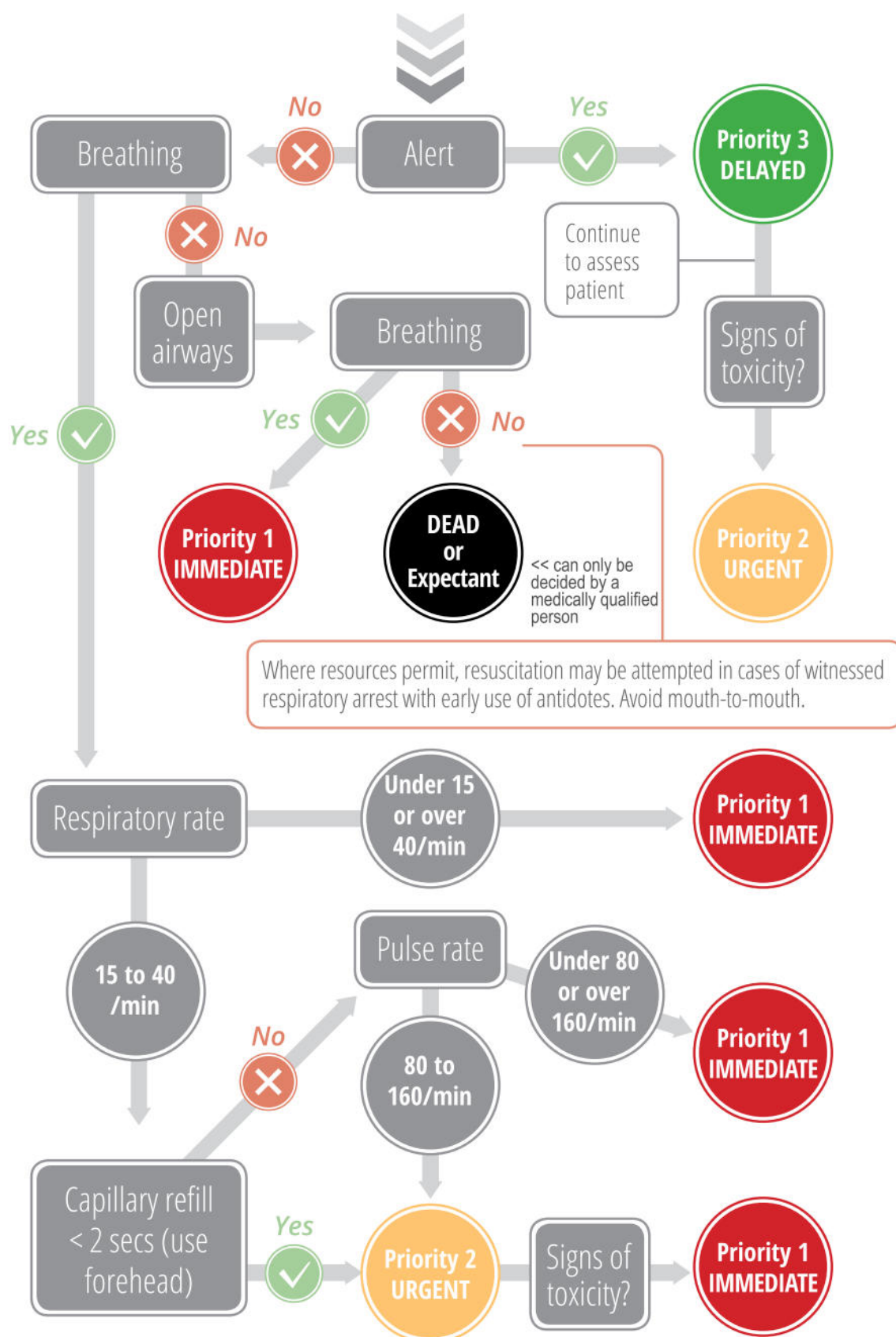
Triage: Children 50-80cm (3-10kg)

Figure 4: Flowchart of the triage of children 3-10kg suspected of exposure to weaponized chemicals



Triage: Children 80-100cm (11-18kg)

Figure 5: Flowchart of the triage of children 11-18kg suspected of exposure to weaponized chemicals



Signs and symptoms of chemical exposures

Table 3: Signs and symptoms of chemical exposures

Class of chemical exposure	Signs and symptoms	
Agent or agent groups		
Nerve agents (e.g. Tabun, Sarin or VX, Novichok ^[7, 12])	Mild	Dizziness, anxiety, headache, weakness Chest tightness, rhinorrhoea, coughing Sweating, salivation, lacrimation Nausea Mild bradycardia and hypotension
	Moderate	Restlessness, confusion, drowsiness, loss of consciousness, muscle twitching, fasciculation Dyspnoea, respiratory depression. Bronchorrhoea and bronchospasm may occur, but were not seen in recent cases of Novichok exposure Miosis/mydriasis*, eye pain Abdominal pain, vomiting, diarrhoea Bradycardia/tachycardia*, hypotension/hypertension*, pallor
	Severe	Convulsions, flaccid paralysis, deep coma Involuntary micturition/defecation
	Fatal	Coma, convulsions Apnoea within a few minutes of exposure Hypersecretions
	<i>*Depending on whether the muscarinic or nicotinic syndrome is dominant, miosis or mydriasis, bradycardia or tachycardia, hypotension or hypertension may occur.</i>	
Blister agents (e.g. mustard gases, lewisite)	Lachrymation, eye irritation, conjunctivitis, corneal damage, transient blindness Hoarse voice, sore throat Delayed signs and symptoms (several hours): Airway irritation, respiratory distress – usually a late complication Redness and blisters of the skin with pain. Later, detachment of the upper skin layers with impaired wound healing Blister formation is typically delayed following exposure to sulfur mustard, but rapid following exposure to lewisite. Immune deficiency	
Cyanide	Gasping for air, asphyxiation, mydriasis Seizures Tachycardia, arrhythmias Confusion Nausea Cherry-pink skin (cyanosis also possible)	
Incapacitating agents		
Agent 15, BZ	Mydriasis Altered consciousness, delusions, hallucinations Ataxia Dry mouth and skin Tachycardia, hyperthermia	
Opioids (e.g. fentanyl analogues, nitazenes) ^[13, 14]	Miosis, drowsiness, dizziness, ataxia, coma, respiratory depression, apnoea	

Signs and symptoms of chemical exposures (continued)

Class of chemical exposure	Signs and symptoms
Riot control agents	
Tear gas/riot control agents (e.g., CS, CR)	Stinging and burning sensation to eyes and mucous membranes Lachrymation/salivation Runny nose Tight chest Headache Nausea
Other toxic chemicals	
Chlorine ^[15]	Eye redness and lachrymation Airway irritation: hoarse voice; cough (may be productive); suffocation or choking sensation, tight chest, shortness of breath/wheezing Nausea and vomiting Delayed signs and symptoms (a few hours): pulmonary oedema
Phosgene	Eye redness and lachrymation Nausea and vomiting Tight chest, shortness of breath/wheezing Hypotension Delayed signs and symptoms (up to 72 hours): pulmonary oedema
White Phosphorus	Severe, painful burns, yellowish necrosis, slow healing Risk of re-ignition by air exposure Corneal burns, irritation, photophobia, conjunctivitis Coughing, lung oedema Delayed signs and symptoms (up to 24 hours): cardiovascular collapse, renal/hepatic failure, CNS depression. After phosphorus penetrates the skin, hyperphosphatemia, hypocalcemia and electrocardiogram changes may occur depending on degree of exposure.

Treatment protocols for weaponized chemical exposures

Tables 4-10 below provide brief information on treatment and cannot address all eventualities.

- The antidote regimens listed below are for guidance, there are a number of regimens in current use by medical authorities in different countries.
- For all exposed patients, perform an 'ABCDE'¹ medical assessment.^[2]
- Provide supportive care, including invasive mechanical ventilation when available, in conjunction with the use of medical countermeasures/antidotes.
- Closely monitor patients, as their condition may deteriorate particularly if complete decontamination was not achieved.
- Be aware of potential for co-existing blast and penetrating injury associated with a violent incident.
- After the initial response, seek expert consultation to address potential complications.

¹ A-Airway; B-Breathing; C-Circulation; D-Disability; E-Exposure and Environment

Table 4: Specific treatment and antidotes (nerve agents)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
NERVE AGENTS (e.g. sarin, GB, VX, tabun, Novichok) ^[7, 9, 10, 16–19]	Atropine	Adult and children 12 years and over Atropine 2 mg IM or IV every 5–10 minutes. For severe symptoms, up to 6 mg can be given on initial dose. Children (<12 years) Atropine 0.05–0.1 mg/kg IM or 0.02 mg/kg IV (not to exceed 2 mg per dose) every 5–10 minutes. Higher doses/rates of atropine may be required but must be titrated against clinical response (e.g., heart rate).	Administer to all moderate and severe cases. Administer oxygen for hypoxia, but do not delay giving atropine.^[20] Assisted ventilation may be required for severe exposure. Careful monitoring is required in atropine use to ensure that an adequate dose is given and to prevent overdose. During atropine therapy, where possible, use continuous ECG monitoring. Do not use pupil size as a guide to adequate atropine administration. Repeated antidote administration may be necessary until “atropinisation” achieved by: 1) Clear chest on auscultation. 2) Heart rate > 80 beats per minute. 3) Dry skin (axillae). Onset of symptoms from dermal contact with chemicals in liquid form may be delayed. Observe contaminated asymptomatic patients. Autoinjector formulations are available for these antidotes. Standard autoinjectors are intended for adults: the needles are too long for the muscle bulk of very young children, and the dose of antidotes is too high. The use of a standard autoinjector could be considered in the case of a severely poisoned child needing urgent treatment, however, the described limitations should be taken into account.^[21]
	Hyoscine hydrobromide (with atropine)	Adult Hyoscine hydrobromide 0.3–0.6 mg IV or IM Child Hyoscine hydrobromide 6 µg/kg (max 3 mg per dose) IV or IM	Hyoscine hydrobromide penetrates the central nervous system more effectively than atropine. It may be used as an adjunct in patients with prominent central features such as agitation, confusion, or persistent hypersalivation. Use in addition to ongoing atropine therapy. Do not delay atropine treatment or dose escalation.

Table 5: Specific treatment and antidotes (nerve agents, continued)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
Nerve agents (e.g. Sarin, GB, VX, tabun, Novichok) ^[16]	Oximes	Adult Pralidoxime (as chloride or mesylate salt): 30 mg/kg (up to 2g) slow IV. Repeat every 4-6 hours, or give infusion of 8-10 mg/kg/hour OR Obidoxime: 250 mg IM or slow IV followed by infusion of 750 mg in 24 hours. Maximal daily dose 1000 mg Child Pralidoxime (as chloride or mesylate salt): 15-30 mg/kg slow IV. Repeat once at 30-60 minutes, then at one hour intervals for 1-2 doses, as necessary. OR Obidoxime: 4-8 mg/kg by slow IV; in case of need followed by infusion with 10 mg/kg/24 hours	To regenerate acetylcholinesterase Other oximes such as HI-6 can be utilized but are not widely available Autoinjector formulations are available for these antidotes. Standard autoinjectors are intended for adults: the needles are too long for the muscle bulk of very young children, and the dose of antidotes is too high. The use of a standard autoinjector could be considered in the case of a severely poisoned child needing urgent treatment, however, the described limitations should be taken into account.^[22] Early administration (within the first 24 hours) is recommended. However, pralidoxime may be beneficial even when given later due to off-target effects such as anti-nicotinic action. Continued pralidoxime may help if there are delayed effects with nerve agents in tissue and where there is slow release.^[23]
	Benzo-diazepines	Adult Diazepam 5-10 mg IV/IM (higher doses up to 40 mg, may be necessary). Child Diazepam 0.05 to 0.3 mg/kg IV/IM (maximum 10 mg).	For treatment of seizures or empirically in severe cases. Other benzodiazepines (e.g. lorazepam, midazolam) can be used.

Table 6: Specific treatment and antidotes (lung irritants)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
Lung irritants (e.g. chlorine, phosgene) ^[15, 24-26]		N-acetylcysteine (NAC) (IV infusion) may be of benefit following exposure to chlorine.^[24] Use for phosgene exposure is maintained in some guidelines, but unlikely to be of benefit.^[24, 27]	Put patient at rest in a semi-upright position and keep warm. Give symptomatic therapy as required e.g. oxygen and bronchodilators^[15] Assisted ventilation may be required. Treat cough with codeine 30-60 mg/day.

Table 7: Specific treatment and antidotes (blister agents)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
Blister agents (e.g. Sulfur mustard, lewisite) [28, 29]	Sodium thiosulfate <i>(for severe exposure)</i>	Adult and child Sodium thiosulfate: 400 mg/kg to max of 12.5g (e.g. 1.6 mL/kg to max 50 mL of 25% solution) over 10 minutes.	If severe exposure to mustard is suspected, consider use of intravenous sodium thiosulfate to decrease systemic effects. This must be done within first hour, and preferably 20 minutes after exposure.
	Supportive care	Eyes Irrigate with distilled water or copious 0.9% sterile saline solution and then use sterile petroleum jelly or ophthalmic ointments, such as 5% boric acid to prevent eyelids sticking together. Prevent infection with a topical antibiotic e.g. Sulfacetamide eye drops. Apply local anaesthetic drops if necessary to relieve blepharospasm and enable eye examination (though anaesthetic drops may increase risk of damage to the cornea). Do not patch the eye. Skin Wash affected skin with copious amounts of soap and water. Itching can be reduced by local applications of cooling preparations, e.g. calamine lotion, or water. Apply silver sulfadiazine cream and cover with sterile dressing. Analgesics should be given as required. Respiratory tract Cough may be relieved by codeine. In severe cases, aggressive airway management and bronchial lavage with mechanical ventilation/positive end expiratory pressure (PEEP) and oxygen administration may be required. Cricothyrotomy rather than endotracheal intubation may be appropriate when there is significant upper airway involvement. Bronchodilators may be useful if bronchospasm is significant. The use of N-acetylcysteine may be of benefit, but it should be given as the intravenous preparation. In the case of ingestion do not induce vomiting.	
	Anti-lewisite (specific antidote)	DMPS (Dimaval®): IV and oral use. Dose regimen in severe poisoning with lewisite: Adults Day 1: 250 mg DMPS IV every 3-4 hours Day 2: 250 mg DMPS IV every 4-6 hours Day 3: 250 mg DMPS IV every 6-8 hours Thereafter: 250 mg DMPS IV every 8-12 hours then 250 mg DMPS 1-3 times per day or switch to oral (100 mg 1-4 times per day).	British anti-lewisite/dimercaprol remains available in some formularies. While it can be effective, it has significant toxicities. Expert advice should be sought regarding its use if DMPS is not available.

Table 8: Specific treatment and antidotes (cyanide)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
Cyanide ^[30]	See below for details. Amyl nitrite can be used as first-aid antidote. Subsequent treatment should use either 1) sodium thiosulfate [which may be combined with sodium nitrite or 4-DMAP] or 2) hydroxocobalamin. Sodium thiosulfate is sometimes used in combination with hydroxocobalamin but this is not essential.		
	Amyl nitrite (<i>first aid</i>)	Crushed 0.3 ml ampoule inhaled for 15 sec, may repeat after 3-5 min. Amyl nitrite ampoules may be broken into an Ambu bag or similar resuscitator if the patient is not breathing.	100% oxygen therapy, ventilate and intubate, if indicated. Patients who reach hospital alive after having inhaled cyanide probably do not need antidotal treatment since cyanide acts quickly. Monitor the patient for metabolic acidosis. Use of 50 – 100 ml of 8.4% solution sodium bicarbonate may be considered. Correct electrolyte imbalance.
	Sodium thiosulfate ^[31]	Adult and child Sodium thiosulfate: 400 mg/kg to max of 12.5g (e.g. 1.6 ml/kg to max 50 ml of 25% solution) over 10 minutes. Additional doses may be given if necessary.	Sodium thiosulfate is a substrate for rhodanese and aids metabolism of cyanide. Sodium thiosulfate is used together with sodium nitrite or 4-DMAP. For mild poisoning, sodium thiosulfate may be used alone. Sodium thiosulfate is sometimes used with hydroxocobalamin but this is not essential.
	Sodium nitrite (<i>combination with sodium thiosulfate</i>)	Adult Sodium nitrite 300 mg (e.g. 10 ml of 3% solution) by slow IV over 5-10 min. Sodium nitrite can be repeated at 50% of the original dose after 30 mins if no improvement. Child Sodium nitrite: 4-10 mg/kg to max of 300 mg by slow IV. (3% solution: 0.13-0.33 ml/kg). Sodium nitrite can be repeated at 50% of the original dose after 30 mins if no improvement.	To induce methaemoglobinemia; not recommended in smoke inhalation Used with sodium thiosulfate. Sodium nitrite may cause haemolysis in patients with G6PD deficiency.
	4-DMAP (<i>combination with sodium thiosulfate</i>)	3-4 mg/kg IV. Single dose. Do not repeat.	To induce methaemoglobinemia; not recommended in smoke inhalation Used with sodium thiosulfate. 4-DMAP may cause haemolysis in patients with G6PD deficiency.
	Hydroxo-cobalamin ^[32]	Adult 5g IV over 15 min Child 70 mg/kg IV over 15 min.	Used for cyanide exposure related to smoke inhalation. ^[33] Safer for patients with hypotension but must use high-dose formulation which is not so widely available. If sodium thiosulfate is also given, do not to administer both drugs through the same IV line.

Table 9: Specific treatment and antidotes (incapacitants)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
Incapacitants BZ, Agent 15	Physostigmine	Adult 2-3 mg IV then maintenance 2-4 mg /hour (caution bradycardia). Aim to switch to oral administration as soon as possible (2-5 mg every 1-2 hours). Therapy should not be stopped too early (4-5 days depending on the severity of poisoning).	Provide symptomatic treatment, with the aim of preventing the patient harming themselves by their actions. Attempt to resolve the situation without physical or chemical restraint. If necessary, however, sedate with a short-acting benzodiazepine (e.g. midazolam).
Incapacitants Opioids (e.g., nitazenes, fentanyl analogues such as carfentanil, remifentanyl and sufentanyl)	Naloxone	Adult Initial dose: 0.4-2 mg IV. If insufficient improvement in respiratory function, repeat every 2-3 mins. If no response is observed after 10 mg of naloxone, reconsider diagnosis of opioid toxicity. Synthetic opioids may require very high doses of naloxone. ^[34] Child Initial dose: 5-10 µg/kg IV. If insufficient clinical improvement, give further dose of 100 µg/kg.	Can also be given IM or subcutaneously if the IV route is not possible, however onset of action will be slower.

Table 10: Specific treatment and antidotes (white phosphorus)

Chemical	Antidotes	Initial treatment	Notes and supportive therapy
White Phosphorus	Clean water or 0.9% sodium chloride	Flush with cool water or saline and remove particles with forceps. Continuous irrigation: keep burn sites continuously wet with saline-soaked dressings; re-irrigate if smoldering resumes. ^[35, 36]	Irrigation prevents burning, but not toxicity.
	Supportive care	Pain management: Administer IV opioids (e.g., morphine, fentanyl) for severe burn pain; titrate to effect. Burn care: Cover burn sites with wet, saline-soaked dressings to prevent re-ignition; maintain moisture throughout care. ^[35, 36] Airway support: Provide high-flow oxygen for smoke inhalation; prepare for intubation if airway burns or respiratory distress are present. Fluid resuscitation: Initiate IV fluids (e.g., lactated Ringer's) for extensive burns or shock; monitor vital signs closely. Cardiac monitoring: Watch for arrhythmias (e.g., QT prolongation) and manage per ACLS protocols; consider IV calcium gluconate for suspected hypocalcemia.	

Methods

Collating clinical best practice guidelines

The contents of the original “*Initial clinical management of patients exposed to chemical weapons: interim guidance document*” were based on a review of existing guidance ^[2, 3, 8, 9, 17, 18, 30, 32, 37–40].

The source documents were identified as a preliminary assessment of selected existing guidance on emergency decontamination, triage and treatment regimens for a range of highly toxic chemicals. Guidance documents were selected based on their availability, either publicly or via WHO Collaborating Centres. The main sources were guidance documents developed by UK and US national institutions. Only English language guidance was consulted. Selected guidance documents were reviewed paying particular attention to their own sources of evidence and their applicability to the contexts of all WHO Member States, including those in resource-limited settings.

The document was developed and reviewed by WHO technical staff and external international experts. Recommendations, and the underpinning evidence were reviewed and agreed at a WHO expert meeting on 9-10 December 2013. The final document was also reviewed by the Global Health Security Action Group (GHSAG) and members of the Chemical Events Working Group.

Update for 2026

General approaches to chemical exposure have not changed, as confirmed by expert review. Links to best practice guidelines (see above) have been verified, including updated versions.^[41]

Systematic review

Rapid systematic review was performed in March 2026 using PubMed using limits from Jan 2014 to March 2026, covering the period following the original document publication ^[42]. Additional search restrictions included limitation to English language, and only full text documents were considered (abstracts not included). A single reviewer screened the results. Results of the searches are given in Table 11; articles which were identified after combination with the antidote terms were screened.

Multiple updated systematic reviews and literature review articles were identified and have been cited within this document against the agents to which they are relevant. Putative treatments, or those which were suggested based on putative benefit extrapolated from mechanistic or animal studies were not included.

Table 11: Rapid systematic review search results

Section	Search terms	# articles (with date restriction)	# articles combined with antidote terms
Cyanide	("Cyanides"[Mesh] OR "Cyanides/poisoning"[Mesh] OR "Cyanides/toxicity"[Mesh] OR "Cyanide poisoning"[tiab] OR "Cyanide toxicity"[tiab] OR "Hydrogen cyanide"[tiab] OR "Prussic acid"[tiab])	271	20
Nerve agent	("Nerve Agents"[Mesh] OR "Chemical Warfare Agents"[Mesh] OR "Sarin"[tiab] OR "Tabun"[tiab] OR "VX agent"[tiab] OR "Novichok"[tiab] OR "A-234"[tiab] OR "A-232"[tiab] OR "A-230"[tiab])	278	66
Blister agents	("Mustard Gas"[Mesh] OR "Mustard Gas/poisoning"[Mesh] OR "Sulfur mustard"[tiab] OR "Bis(2-chloroethyl) sulfide"[tiab] OR "Yperite"[tiab] OR "Mustard agent"[tiab] OR "Lewisite"[tiab] OR "2-chlorovinylchloroarsine"[tiab] OR "Dichloro(2-chlorovinyl)arsine"[tiab])	125	12
Incapacitating agents	("Quinuclidinyl Benzilate"[Mesh] OR "Quinuclidinyl Benzilate"[tiab] OR "BZ agent"[tiab] OR "Agent 15"[tiab] OR "QNB"[tiab] OR "Incapacitating agent"[tiab])	74	0
	("Fentanyl"[Mesh] OR "Benzimidazoles"[Mesh] OR "Analgesics, Opioid"[Mesh] OR "fentanyl"[tiab] OR "carfentanil"[tiab] OR "nitazene"[tiab] OR "isotonitazene"[tiab] OR "synthetic opioid"[tiab])	68731	369
Tear gas/riot control agents	("o-Chlorobenzylidenemalonitrile"[Mesh] OR "Riot Control Agents, Chemical"[Mesh] OR "Tear Gases"[Mesh] OR "CS gas"[tiab] OR "Tear gas"[tiab] OR "2-chlorobenzalmalononitrile"[tiab])	98	1
Chlorine	("Chlorine"[Mesh] OR "Chlorine/poisoning"[Mesh] OR "Chlorine/toxicity"[Mesh] OR "Chlorine gas"[tiab] OR "Chlorine inhalation"[tiab])	3663	15
Phosgene	("Phosgene"[Mesh] OR "Phosgene/poisoning"[Mesh] OR "Phosgene/toxicity"[Mesh] OR "Phosgene"[tiab] OR "carbonyl chloride"[tiab] OR "Phosgene exposure"[tiab])	397	13
White phosphorus	((("Phosphorus"[Mesh] AND ("poisoning"[Subheading] OR "adverse effects"[Subheading] OR "burns"[tiab])) OR "white phosphorus"[tiab] OR "yellow phosphorus"[tiab] OR "phosphorus burns"[tiab])	1209	6
Combination	Combination search terms (Boolean AND with each of the above section search terms)		
Antidotes/medical therapy	("Antidotes"[Mesh] OR "Antidotes"[tiab] OR "Medical countermeasures"[tiab] OR "Antidotal therapy"[tiab] OR "Emergency Treatment"[Mesh] OR "Reversal agents"[tiab] OR "Pharmacotherapy"[tiab])	80595	N/A

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